

# MEMORANDUM

Bernie

To: RA, JGC, PJR  
From: B. Cosell and B. Barker  
Subject: PDP-1 Interface to the Network  
Date: 29 November 1971

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We have made a preliminary pass at defining the PDP-1 interface to the network. The interface will be full duplex and will transfer a word-at-a-time through the I/O register. The interface will take 16 bits at a time from the network and will put them in the low order 16 bits of the PDP-1 word leaving the two high order bits 0. Similarly, on output it will transfer the low order 16 bits of the I/O to the network, ignoring the two high order bits. The interface will have the ability to interrupt the processor either when the 16th bit of a word comes over from the network, or when it has just shifted the 16th bit of a word out to the network; that is, when the input shifter becomes full, or when the output shifter becomes empty. Both interrupts are capable of being independently enabled or disabled by the CPU, but will come in on the same channel. We haven't yet selected an interrupt channel or an IOT code for the interface.

## INPUT SIDE

There will be 7 IOTs dealing with accepting input from the network:

1. SKIP IF WORD DONE - This will skip if the shifter has a complete word ready to be transferred to the PDP-1's I/O.
2. INPUT A WORD - This will jam the current state of the input shifter into the PDP-1's I/O. Note that if the shifter was in the middle of assembling a word from the network, garbage will be transferred into the I/O.
3. SKIP ON NO ERROR - This IOT will skip if the IMP's ready line has not been down since the last time a CLEAR ERROR was done.
4. CLEAR ERROR - This will clear the flip-flop which captures the IMP's ready line being down.

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5. SKIP IF IMP READY — This IOT will skip if the IMP's ready line is up.
6. SKIP IF FINAL INPUT WORD — This IOT will skip if the last bit that was shifted into the interface had "Last Bit" set along with it. Note that this must be done before the word in question is transferred into the I/O.
7. ACTIVATE INPUT INTERRUPT — This will cause an interrupt to be generated the next time the input shifter is full and then will clear the interrupt-activated flop. Note that this may cause an immediate interrupt.

OUTPUT SIDE

There are 6 IOTs for transferring data to the network:

1. SKIP IF OUTPUT SHIFTER EMPTY — This IOT will skip if the interface is able to accept another word from the PDP-1 to be shifted out to the network.
2. OUTPUT WORD — This will jam the I/O into the output shifter. Note that this can give highly confusing results if the shifter is not empty when this IOT is done.
3. FINAL OUT — Jams a word out to the shifter (as in 2, above) but will set the last bit indicator with the low order bit of the word.
4. PUT 1 UP — This will bring up the PDP-1's ready line.
5. PUT 1 DOWN — This will take the PDP-1's ready line down.
6. ACTIVATE OUTPUT INTERRUPT — This will cause the CPU to get interrupted the next time the output shifter is empty and then will clear the interrupt activated flop. Note that this may cause an immediate interrupt.

IoT-402

